25900745 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBIF100 - FUNDAMENTAL IT FOR BIOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** Questions

Each Question carries **2** marks

1. List two common network protocols. [K] / [CO2]
2. Identify three common input devices. [K] / [CO1]
3. Differentiate between hardware and software. [U] / [CO1]
4. Define XML. [K] / [CO3]
5. Name the four main categories of operators in JavaScript. [K] / [CO3]
6. Outline the syntax for a for loop in JavaScript. [K] / [CO3]
7. outline basic application of bioinformatics in agriculture. [K] / [CO4]

[5x2 = 10]

Part B

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

8. Explain the various network topologies. [K] / [CO2]

9. Explain the concept of multitasking and how it is implemented in different operating systems. [U] / [CO1]

10. Define an interpreter and explain its main purpose. [U] / [CO3]

11. List some common CSS properties for text formatting [U] / [CO3]

12. List out basic terminologies in bioinformatics [K] / [CO4]

13. Discuss major milestone in human genome project. [U] / [CO4]

[4x5 = 20]

Part C

Long Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

14. Discuss the different types of IP addresses (IPv4, IPv6) and their implications. [K] / [CO2]

15. Differentiate between high level and low level languages in detail. [U] / [CO3]

16. Show how to create internal and external links using the href attribute. [U] / [CO3]

17. Explain why bioinformatics is known as an interdisciplinary field. [U] / [CO4]

[2x10 = 20]